

Benjamin
Karachun

Biotechnology & Food
Engineering
Full-Stack Developer
AI Utilization

kbeniamin@campus.technion.ac.il
+972 54-612-3730
Israel
LinkedIn ↗
GitHub ↗

00 PROFILE

Technion Biotechnology student bridging life sciences and software engineering. I combine wet-lab proficiency with full-stack development in React, Python, and Firebase, with a focus on computational biology and practical tools that solve real academic and research problems.

01 SELECTED PROJECTS

MathPaster Founder & Developer [mathpaster.com ↗](#)
A visual math editor that turns intuitive equation input into clean LaTeX for AI chats. Includes real-time WYSIWYG rendering, autocomplete, symbol palettes, matrix tools, and a keyboard-triggered browser overlay that keeps equations local and private.

BiotechFood Prep Platform Founder & Developer [bfe.technionprep.com ↗](#)
Architected and deployed an educational platform for Technion students, centralizing semester-by-semester course resources and interactive study tools in a fast React-based experience.

Shoresh Nutrition App Founder & Developer [Open project ↗](#)
A data-driven nutrition calculator and label creator using Israeli Ministry of Health data. Built a Python ETL pipeline and browser-based SQLite workflow for food search, comparisons, recipes, daily calculations, and Israeli-compliant labels.

AlgoPrep Founder & Lead Educator [algotprep.com ↗](#)
Founded a coding education brand with thousands of TikTok followers and converted educational content into dozens of private students. Content focuses on algorithm analysis and developer efficiency.

02 EDUCATION

**Technion — Israel
Institute of Technology**
B.Sc. Biotechnology & Food Engineering
Expected 2027 · Haifa, Israel
Computational Biotechnology, Food Engineering
& Health

Mofet Excellence Program
Bagrut
5 units: Chemistry, Physics, Engineering
Sciences, Math & English

03 CORE STACK

- PythonJavaScriptTypeScript
- SQLReactFirebase
- WebAssemblyHTML/CSSGit
- DockerLinux/Bash
- Chrome Extensions

04 LANGUAGES

Hebrew	Native
English	Fluent
Russian	Fluent

05

EXPERIENCE

Python Tutor Technion Student Community

- Mentored undergraduate students in the Technion's Python programming course.
- Architected and deployed **Python TechnionPrep** ↗, a dedicated platform centralizing study materials and improving the learning workflow.

Digital Transformation Consultant Hanan Shemesh Accounting

- Advised on and implemented technology modernization for an accounting firm.
- Integrated the WhatsApp Business API to automate client communication and request handling.
- Led the digitalization of legacy paper records, improving accessibility and security.

Web Maintenance & Development Aviator.co.il

- Managed and maintained WordPress infrastructure for a commercial client.
- Handled updates, asset management, troubleshooting, and day-to-day reliability.

06

AI & LLM ENGINEERING

Prompt architecture

Reusable prompt libraries for educational content, strict JSON schemas, RTL/LTR content, and MathJax 3 output.

Integration

Applications and workflows built around OpenAI, Gemini, and Claude APIs, plus LangChain concepts.

Agentic workflows

Advanced AI-assisted development, Linux automation, research workflows, Antigravity, and OpenClaw.

Long-term practice

Hands-on experience with language models since GPT-3, spanning prompting, product design, and evaluation.

07

SOFTWARE & DATA

LANGUAGES

Python, JavaScript, TypeScript, SQL

DATA

NumPy, Pandas, SQLite, ETL pipelines

WEB

React, HTML5/CSS3, Firebase, WebAssembly

PLATFORMS

GitHub Pages, Chrome Extensions, WordPress

TOOLS

Git, Linux/Bash, Docker, SSH

08

LABORATORY

MOLECULAR

PCR, gel electrophoresis

MICROBIOLOGY

Aseptic technique, pour plate, spread plate, Drigalski method

INSTRUMENTATION

Stomacher, 96-well plate readers, spectrophotometry

BUILDING AT THE
INTERSECTION OF

Science
Education
Software

Explore more on GitHub ↗